

Do **not** write solutions on this page.

12. [Maximum mark: 21]

A curve C has equation $y = \frac{2x^2 + 6x - 3}{x + k}$, $x \in \mathbb{R}$, $x \neq -k$, where k is a real positive constant.

(a) Show that $\frac{dy}{dx} = \frac{2x^2 + 4kx + 6k + 3}{(x + k)^2}$. [4]

(b) Find the range of values of k for which a local minimum or maximum point exists. [4]

Consider the curve C , when $k = 2$.

(c) Write down the equation of the vertical asymptote. [1]

(d) Find the equation of the oblique asymptote. [4]

(e) Show that $\frac{dy}{dx} > 2$, for $x \in \mathbb{R}$, $x \neq -2$. [4]

(f) Sketch the curve C , showing clearly both asymptotes and the general behaviour of C as it approaches each asymptote. [You are not required to find any axes intercepts.] [4]

